

R2 Body Controller | PCA Channel Wiring List

Both PCA9685 boards - all 32 assigned outputs - ESP32-C3 Super Mini body controller

PCA #1 | I2C 0x40

A0-A5 open

Local channel	Board 0-15	Label	Connect to
1	0	FL	Front-left large door
2	1	FR	Front-right large door
3	2	BL	Back-left large door
4	3	BR	Back-right large door
5	4	CB	Charging-bay door
6	5	DP	Data-port door
7	6	SD	Small / short breadpan door
8	7	UA	Upper utility arm
9	8	LA	Lower utility arm
10	9	FLL	Front-left service arm lift
11	10	FLA	Front-left service arm animation
12	11	FRL	Front-right service arm lift
13	12	FRA	Front-right service arm animation
14	13	BZD	Buzzsaw deploy (continuous servo)
15	14	BZM	Buzzsaw motor control signal
16	15	TCM	Trading-card dispenser motor signal

PCA #2 | I2C 0x41

Bridge A0 only; leave A1-A5 open

Local channel	Board 0-15	Label	Connect to
1	0	D1	Drawer 1
2	1	D2	Drawer 2
3	2	D3	Drawer 3
4	3	D4	Drawer 4
5	4	D5	Drawer 5
6	5	D6	Drawer 6
7	6	D7	Drawer 7
8	7	D8	Drawer 8
9	8	BLL	Back-left service arm lift
10	9	BLA	Back-left service arm animation
11	10	BRL	Back-right service arm lift
12	11	BRA	Back-right service arm animation
13	12	TCE	TCD cartridge-eject servo
14	13	X1	User-configurable accessory 1
15	14	X2	User-configurable accessory 2
16	15	X3	User-configurable accessory 3

Numbering: Local channel 1 = board position 0; local channel 16 = board position 15.

These are connector assignments, not BT command indices. Global channels: PCA #1 = 1-16; PCA #2 = 17-32.

X1-X3 are builder-configurable outputs. Service arms are identified by location, regardless of installed arm type.